Juicetification: Aggregate Anxiety

Build a 12-month production plan for a juice bottler — student quick-start

Your mission

You are the operations analyst at Juicetification Inc. Demand peaks in spring and summer. Build a **chase** plan and a **level** plan yourself, compare them, recommend one in your own words, then **download a PDF report and upload it to your LMS assignment**. Everyone gets different numbers — the method is the same.

The roadmap

5 steps in the sidebar	What you do
1 · Understand the scenario	Read the brief and frame the trade-off.
2 · Prepare the worksheet	Pick the data and columns, choose formulas, work out capacity.
3 · Build chase and level plans	Fill the chase and level worksheets in the app — or in Excel, then upload.
4 · Compare and improve	Read the metrics side by side; optional hybrid and advanced exercises.
5 · Submit and reflect	Write your recommendation and reflections; download the PDF.

Every stage shows **your task**, **what to do now**, your progress, and a **Check my work** button. Feedback appears when you press it, so you attempt first. If a stage is incomplete, the page tells you why. Your work is kept as you move around.

Using the worksheet

• Type a number or a formula starting with **=**, e.g. **=E2/RATE** or **=MAX(0,C2+E2-D2-B2)**.
• **Row 1 holds the headings; January is row 2**; column A = Month, B = Forecast Demand — exactly as in the Excel template.
• Functions: SUM, MAX, MIN, AVERAGE, COUNT, ROUND, ROUNDUP, ROUNDDOWN, ABS, IF. Comparisons = < > <= >= <> work inside IF. Powers use ^ . Names: LABOR, HIRE, HOLD, RATE, BEGIN, SAFETY …
• Status always has a symbol: ✔ correct · ✖ needs attention · ○ blank · ⚠ formula error (hover for the reason).
• Prefer Excel? Download the workbook for your scenario, fill it in, upload it; it is checked the same way.
• Enter the plan's total annual cost (sum of Total Monthly Cost) to finish each plan.

Key formulas (defaults — your brief shows your numbers)

• Capacity per worker = 5 /h × 8 h × 25 d = 1,000/month (you are paid for capacity even if you use less)
• Inventory and backlog are separate: physical stock is never negative; unfilled orders are backlog that carries forward, oldest first
• Ending Inventory = MAX(0, Begin Inv + Production − Begin Backlog − Demand)
• Backorders = MAX(0, Demand + Begin Backlog − Begin Inv − Production)
• Workers (whole) = ROUNDUP(Production ÷ capacity) · Paid Labor Hours/Day = Workers × hours per shift
• Hires = MAX(0, this month − last month) · Layoffs = MAX(0, last month − this month)
• Level rate = (Total demand + ending target − Beginning inventory) ÷ 12
• Safety stock is a **planning target**, not a hard floor: stock below it is still shipped and flagged
• Total Monthly Cost = Labor + Hiring + Layoff + Holding + Backorder

Default cost sheet (your instructor may change these)

Labor $3,200/worker/mo · Hire $600 · Layoff $900 · Holding $0.25 · Backorder $1.50 · Overtime $4.50 (≤20%) · Subcontract $5.25 · Begin 2,400 · Start workforce 8

Stuck?

Each wrong cell is explained. “I'm stuck” (if your instructor allows it) shows a worked solution and then gives you a **replacement scenario** — different numbers — so you rebuild it yourself. Help use is recorded neutrally in your report.

Submitting

When every required activity is complete, press **Generate final report**, then **Download PDF for LMS submission**. Download your report, review it, and upload the PDF to your LMS assignment. A draft can be downloaded at any time but every page is marked DRAFT. The cost shown in your report is **plan cost, not a grade**.

Run locally: streamlit run app.py · model 3.0.0